PAPER REVIEW NR 4

Paper:

Mengyao Li & John D. Lee (2026) Purpose Outweighs Performance: Trust Drops More When AI Teammates Fail to Cooperate, but Explanations Can Repair It, International Journal of Human–Computer Interaction, 42:6, 4378-4399, DOI: 10.1080/10447318.2025.2539449

Reviewer:

Lukas Steu

I confirm that I have read the paper and written the following texts myself ☒

1. Thematic focus

This paper studies trust in human-AI teams. It compares performance-based and purpose-based trust violations and it also examines which repair strategies can restore trust after these violations.

2. Foundations

- Trust in automation by Lee and See
- Human-AI teaming and human-autonomy teams
- Game theory, especially the Trust Game and Threshold Public Goods Game
- Trust repair theory by Pak and Rovira
- Multi-Dimensional Measure of Trust by Ullman and Malle

3. Method

This is a mixed-factorial desing study with a 2x2x3 framework. The AI has the states low and high, then the violation states differ between purpose or performance and the trust repair strategies differ in no repair, apology + explanation or apology + promise.

4. Key results

- Purpose-based trust violations reduced trust more strongly than performance-based violations
- Performance-based violations did not significantly reduce trust when the AI still cooperated with the team goal
- An apology combined with an explanation repaired trust after purpose-based violations
- Promises after purpose-based violations could even reduce cooperation.

5. Practical implications for AI or robotics

AI systems should not only be reliable but also share the same goals with human and the team. After a mistake, AI should provide clear explanations instead of only making promises.

Designers should be careful with promises, because broken promises can damage cooperation even further then the damage already dealt.

6. Strengths of the paper

The paper makes a clear distinction between performance-based and purpose-based trust. The study combines behavioural measures with subjective trust ratings, which makes the results stronger compared to other studies.

7. Weaknesses of the paper

The study was conducted online, so the researchers could not observe participants in detail. Only two trust violations were used, so the results may differ when violations happen more often. The AI messages were quite simple and may not fully represent real human-AI communication.

8. Personal learnings

Watch out for future promises AI models send you! Just kidding, but don't get too dependent on AI so you don't have to trust it entirely 😊.